KIQSTAT · ASK KIQSTAT · 29 OCTOBER 2026

Staging verification and margin floor

Migration 0088 was verified on a development database, and the worst-case cost of every paid option is now bounded and priced. The live benchmark hasn't run yet: it needs a test key that only you can create. Nothing in production has changed.

MEASURED from a test run, the database, Vercel billing or the code

ESTIMATED from prompt sizes and published prices, until the benchmark runs

VERIFIED

Migration 0088 applies cleanly after the other 87 migrations on a development Supabase stack. The logging, reservation, settlement and alert code passed 9 of 9 checks through the real API, including 40 parallel requests from one account. MEASURED

BLOCKED

The live benchmark (5 settings × 40 questions) and the tests against Anthropic itself need `ANTHROPIC_TEST_API_KEY`. It doesn't exist yet, and creating workspaces and keys happens in your Anthropic console.

EXPOSURE

Until the guard is switched to enforce, nothing caps Ask's spend. A VIP member asking all 100 questions a day at the worst question cost could cost about ₦164,000 in 31 days, against a ₦12,000 price. ESTIMATED

BELOW 25%

At today's caps, five paid options fall below 25% in the worst case: two Paystack card plans that charge less than the site shows, the yearly VIP plan, and all three passes. One reason is that a 24-hour pass spans two Lagos days, so it gets two days of cap. MEASURED prices and fees, worst-case AI by cap

FIX

With the caps per option proposed here, every paid option clears 30% at ₦1,600 to the dollar and stays at or above 25% up to about ₦1,760. However much a member asks, the cap holds, so this floor doesn't depend on the benchmark.

SCALE

Vercel costs about ₦90,000 a month, measured. October's payments came to ₦72,250. Fixed infrastructure is more than today's revenue,

whatever the AI settings.

1. Where each part of the assignment stands

STEP	STATUS	WHAT HAPPENED
1. 0088 on staging first	DONE	Supabase branching needs the Pro plan; the organisation is on Free. Creating a second hosted project was blocked here, because it's a billing action on your account. Instead I ran the full Supabase stack (Postgres 17, PostgREST, auth) locally and applied all 88 migrations. `scripts/ask-staging.sh` repeats it in one command.
2. Separate capped Anthropic workspaces	NEEDS YOU	Only `ANTHROPIC_API_KEY` exists in Vercel. Steps are in section 6. I won't handle key values.
3. Five settings × 40 questions	READY, NOT RUN	The lab (`/admin/ask-lab`) runs it on the Preview deployment once the test key is there. It records cost, latency, completeness and ratings per answer.
4. Worst case against the provider	HALF DONE	Concurrency, cancellation, budget exhaustion and the 31-day exposure were proven on the database and in the engine (section 3). Repeating them against Anthropic itself needs the test key.
5. Margin per billing period	DONE	Measured prices, fees, commission and Vercel costs, with worst-case AI bounded by the caps (section 5).

2. Staging verification of 0088

These checks run through the same `reserveSpend` , `settleSpend` , `logQuestion` and `checkAlerts` code that the live route calls, with supabase-js going through PostgREST, as in production. MEASURED , 9 of 9 passed (`src/lib/ask/ledger.staging.test.ts`).

CHECK	RESULT
Browsers can't read the books	The anon key gets refused on all five tables and on `ask_spend_reserve` .
Spending is booked to the Lagos day	Both holds land on the Africa/Lagos date.
Pro day cap ($0.04)	Two $0.02 questions are granted and the third is refused (`subject`). Settling to $0.0021 frees the rest of the day's room.
40 parallel requests, one account	Exactly 5 granted, totalling exactly the $0.10 cap. Spending every grant lands the account on the cap, and the next request is refused.

Near the cap	With $0.005 left, it grants $0.005, enough for a short answer. With $0.0001 left, it refuses.
Observe mode	Every request goes through, and the ones past the cap are flagged as `subject`.
Cancellation	A stream aborted after 500 output tokens is billed for those 500. The question is logged as `aborted`, with its call marked as cancelled.
Budget exhaustion	Four VIPs in parallel against a $0.03 budget: no more than $0.03 is granted, and the next request is refused (`budget`). Each alert email is sent once, not on every request.
Books unreachable	Enforce mode refuses the question; observe mode lets it through.

Schema drift found: migration 0015 revokes `public.rls_auto_enable()`, a function no migration creates; production got it from outside the migrations. A fresh database fails at 0015 without it. The staging script adds a stub; a one-line migration (or a guarded revoke) would let a clean database build on its own.

3. Worst-case cost evidence

PATH TO OVERSPEND	WHAT STOPS IT	EVIDENCE
One long question	Before each call, the engine checks what that call could cost at most (its whole prompt at the cache-write rate plus every output token allowed) and shortens the call or stops.	200 random trials never passed the ceiling. MEASURED
Parallel requests	One locked row per account and day; each request is granted only what is left.	40 at once: total granted = the cap exactly. MEASURED
Cancelling to dodge the bill	Usage is read from the stream as it is cancelled, then settled.	The tokens streamed before the cancel are billed and logged. MEASURED
Retries	The SDK retries a refused or overloaded request twice; Anthropic doesn't bill those refused attempts. The engine doesn't retry after an API error. A malformed tool call is re-issued at most twice, inside the same reservation.	From the code and the engine tests. MEASURED. To be confirmed against the API with the test key.
Everyone at once	A $5 daily budget across all of Ask, reserved before any account cap.	Exhaustion refuses requests and alerts once. MEASURED

31-day exposure per member

Worst question under today's production settings (8,000 tokens a call, 6 calls) is about ₦53. ESTIMATED.

OPTION	PRICE	TODAY: GUARD OBSERVING, NO CAP	CAPS ENFORCED (PROPOSED)
Pro, 30 questions a day, 31 days	₦5,000	≈ ₦49,300	≤ ₦1,984
VIP, 100 questions a day, 31 days	₦12,000	≈ ₦164,300	≤ ₦4,960
VIP day pass, 24 hours across two Lagos days	₦500	≈ ₦10,600	≤ ₦224 (per-pass cap)

The uncapped figures only become real once Anthropic has credit again; Ask has been refused for credit since about 10 October. That's why the guard should go to enforce in the same change that restores credit.

4. Infrastructure, measured

Vercel charges for 20–27 October averaged **$1.86 a day, about $56 a month (₦90,200)**. Of the daily total, the Pro seat is $0.65, observability $0.37, function CPU and memory $0.35, builds $0.19 and invocations $0.09. Supabase is on the Free plan ($0). MEASURED. October's successful payments came to ₦72,250 across 12 payments. Active paid accounts: 8 Pro, 4 VIP, 4 passes. MEASURED

Fixed infrastructure doesn't grow with each member, so the margin table charges it separately. Each option is charged for its Ask function time instead (worst case: 15 seconds at 2 GB per question). For fixed costs to take no more than 5 points of margin, revenue needs to be about ₦1.8 million a month, roughly 360 Pro-monthly members.

5. Contribution margin per billing option

Revenue minus the Paystack fee (1.5% plus ₦100 from ₦2,500, capped at ₦2,000) and a 20% partner commission on the net, as if every sale were referred. Then minus worst-case AI, meaning the member uses the whole cap every day of the period (31, 92 or 366 days), and Ask function time. At ₦1,600 to the dollar. Green ≥ 30%, amber 25–30%, red under 25%.

At today's caps

OPTION	PRICE	DAYS	PAYSTACK FEE	COMMISSION	CAP NOW	AI WORST CASE	MARGIN
Pro monthly	₦5,000	31	₦175	₦965	$0.04/day	₦1,984	33.8%
Pro 3 months (site / pay-once)	₦13,500	92	₦302	₦2,640	$0.04/day	₦5,888	30.6%
Pro 3 months (Paystack card	₦9,500	92	₦242	₦1,852	$0.04/day	₦5,888	10.2%

plan)							
Pro yearly	₦48,000	366	₦820	₦9,436	$0.04/day	₦23,424	25.3%
VIP monthly	₦12,000	31	₦280	₦2,344	$0.10/day	₦4,960	31.7%
VIP 3 months	₦32,400	92	₦586	₦6,363	$0.10/day	₦14,720	27.5%
VIP yearly (site / pay-once)	₦115,200	366	₦1,828	₦22,674	$0.10/day	₦58,560	21.6%
VIP yearly (Paystack card plan)	₦76,800	366	₦1,252	₦15,110	$0.10/day	₦58,560	-7.0%
VIP day pass, 24 h (active)	₦500	1	₦8	₦98	$0.12/day over 2 Lagos days	₦384	-5.9%
VIP weekend, 72 h (draft)	₦1,000	3	₦15	₦197	$0.12/day over 4 Lagos days	₦768	-5.9%
VIP trial, 24 h (draft)	₦300	1	₦4	₦59	$0.12/day over 2 Lagos days	₦384	-62.4%

With the proposed caps

OPTION	CAP PROPOSED	AI WORST CASE	FUNCTION TIME	CONTRIBUTION	MARGIN	AT ₦1,900/$
Pro monthly	$0.040/day	₦1,984	₦184	₦1,692	33.8%	25.7%
Pro 3 months (site / pay-once)	$0.040/day	₦5,888	₦545	₦4,125	30.6%	21.6%
Pro 3 months (Paystack card plan)	$0.025/day	₦3,680	₦545	₦3,181	33.5%	25.1%
Pro yearly	$0.035/day	₦20,496	₦2,167	₦15,081	31.4%	22.6%
VIP monthly	$0.100/day	₦4,960	₦612	₦3,804	31.7%	23.0%
VIP 3 months	$0.090/day	₦13,248	₦1,815	₦10,388	32.1%	23.3%
VIP yearly (site / pay-once)	$0.080/day	₦46,848	₦7,222	₦36,627	31.8%	23.0%
VIP yearly (Paystack card plan)	$0.050/day	₦29,280	₦7,222	₦23,936	31.2%	22.3%
VIP day pass, 24 h (active)	$0.140 per pass	₦224	₦20	₦150	30.1%	20.9%
VIP weekend, 72 h (draft)	$0.255 per pass	₦408	₦59	₦321	32.1%	23.3%
VIP trial, 24 h (draft)	$0.075 per pass	₦120	₦20	₦97	32.2%	23.5%

The Paystack card plans charge ₦9,500 for Pro 3 months and ₦76,800 for VIP yearly; the site shows ₦13,500 and ₦115,200. Payments at ₦9,500 appear in the records. The better fix is to correct the two Paystack plans; the lower caps are only needed if those prices stay. Pass caps apply over the whole pass rather than per Lagos day. That needs a small code change, because spending is booked per day today.

What the proposed caps mean for members, at about ₦2 a typical question (ESTIMATED, to be replaced by the benchmark): Pro about 28–32 questions a day before the cap, so the

allowance of 30 holds. VIP about 64–80, against an allowance of 100. Day pass about 112. If the benchmark shows typical questions cost less, VIP gets nearer 100 with no change to the cap.

6. What I need from you

1. **Anthropic console:** create two workspaces, *Ask KiqStat* and *Ask lab*, each with a monthly spend limit. Suggested limits: $150 for Ask (30 days of the $5 budget) and $10 for the lab, which is enough for several full runs. Create one key in each.
2. **Vercel:** add the lab key as `ANTHROPIC_TEST_API_KEY` on **Preview only**. Add the Ask key as `ANTHROPIC_ASK_API_KEY` once you're ready to move production onto its own workspace. Production keeps using the current key until then.
3. **Run the benchmark:** open `/admin/ask-lab` on the latest dev Preview, signed in as an admin. Run each of the five settings and rate the answers. I'll then turn the measured costs, latency and ratings into the final table. Or give this environment a lab key and I'll run it and repeat section 3 against Anthropic.
4. **Decide:** fix the two Paystack plan prices, adopt the per-option caps, and switch `ASK_GUARD` to `enforce` together with the credit top-up.